Premier University Chittagong
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EpiScope

A Scalable District-Level Disease Surveillance and Response Platform for Bangladesh

Complex Engineering Project (CEP) Technical Report

CSE4345 – Big Data Analytics

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Submission Date: 02-11-2026

EpiScope: A Scalable District-Level Disease Surveillance and Response Platform for Bangladesh

CSE4345 Big Data Analytics – CEP Technical Report

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***Abstract*—District health offices in Bangladesh must act on dengue, diarrhoeal disease, influenza-like illness, and malaria using data that arrive late, incompletely, and in separate systems. This report presents EpiScope, a reproducible district-level surveillance and response prototype that integrates a synthetic five-year encounter corpus, a PySpark Bronze–Silver–Gold pipeline, four analytical components, and a static D3 dashboard. The pipeline ingests 6,037,903 synthetic encounters for 64 districts and five disease groups, removes or quarantines 150,794 records (2.5%) under twelve validation controls, replaces patient identifiers with a salted SHA-256 hash, and publishes 83,127 district–disease–week records with small-cell suppression. A weekly adaptation of the EARS C2 detector scores 76,535 series-weeks and raises 4,806 high-level signals; SARIMA models produce 1,280 forecast rows across 320 series; a linear program allocates 19,806 of 33,010 requested units; and a tercile calendar labels 16,938 district–disease–weeks. Against 30 district-level withheld synthetic outbreaks, 26 (87%) produced at least one signal within the labeled window plus or minus seven days, with a median first-signal offset of five days. A rolling-origin backtest of the 320 forecasting series over the last 52 weeks reduces MAE by 22.9% against a seasonal-naive baseline, with 87.5% realised coverage of the 80% prediction intervals. The report also maps the work to the BAETE complex-engineering-problem attributes P1–P7, resolves the five required requirement conflicts with measurements, and discusses ethical, societal, and sustainability considerations. EpiScope supports human review rather than automatic outbreak declaration.**

***Index Terms*—Apache Spark, Bangladesh, disease surveillance, early aberration detection, linear programming, public-health dashboard, SARIMA.**

I. Introduction and Problem Analysis

A. Problem

Bangladesh faces a seasonal and geographically uneven infectious-disease burden. Dengue clusters in dense urban districts during the monsoon, diarrhoeal disease rises with flooding and temperature, influenza-like illness (ILI) increases in winter, and malaria persists in the south-east. District health managers must decide where to intensify surveillance, how many beds and intravenous fluids to pre-position, and which signals warrant field investigation. These decisions require joining case reports, population denominators, weather, facility capacity, and stock data that are usually managed in separate systems. Delayed reporting and inconsistent case definitions further compress the time available for response.

The stakeholders and their conflicting needs were analysed at the proposal stage and are summarised in Table I. The Ministry of Health needs a defensible national picture; epidemiologists need transparent, adjustable methods; district officers need early local warning without alert fatigue; hospital managers need forecasts with uncertainty; and privacy regulators require that no line-level data leave secure zones. EpiScope was designed around these tensions rather than around a single technical showcase.

B. What EpiScope Does

EpiScope is a district-level prototype that turns routine-like encounter data into (i) early-warning signals, (ii) one-to-four-week planning forecasts, (iii) supply-allocation recommendations under a fixed national budget, and (iv) a seasonal preparedness calendar, all exposed through a plain-language static dashboard. It is deliberately decision-support: signals and forecasts prompt human review, and allocation outputs are recommendations rather than instructions. International frameworks motivate the design: countries must detect and respond to public-health events under the International Health Regulations [1], dengue remains a major seasonal threat [2], and the WHO Early Warning, Alert and Re-

TABLE I
STAKEHOLDERS AND THEIR PRIMARY NEEDS AND CONCERNS.

Stakeholder	Primary need	Typical concern
MoH policymakers	National picture; defensible allocation	"Why did district X get more than Y?"
Surveillance epidemiologists	Reliable, transparent signals	Artefacts treated as outbreaks
District health officers	Early local warning and forecast	Alert fatigue; delayed supplies
Hospital managers	Demand forecasts with uncertainty	Forecasts without intervals
Vector-control units	Hotspots and timing	Map too coarse
Patients and public	Timely response; no data misuse	Re-identification; stigma
Privacy regulator	Minimal data; suppressed small counts	Line-level data exposed

sponse System (EWARS) offers a lightweight workflow for outbreak-prone settings [4]. Public-health surveillance itself is the systematic collection, analysis, and dissemination of health data for action [3].

C. Contributions and Report Structure

The contributions are: a reproducible PySpark pipeline with retained quarantine and batch lineage; a governance layer of twelve validation controls, salted hashing, and small-cell suppression; an integrated analytical stack (weekly EARS C2 with a CUSUM comparison, SARIMA with intervals, linear-programming allocation, and a tercile calendar); a responsive D3 dashboard; and an evaluation with explicit limitations. Section III maps the solution to the complex-engineering-problem attributes and resolves the five required conflicts. Section IV presents the architecture and design decisions with alternatives. Section V covers data generation, quality, lineage, and privacy. Section VI describes the analytics. Section VII evaluates the system. Section VIII discusses the dashboard and usability findings. Sections IX and X cover ethics and limitations.

II. Related Work

A. Surveillance and Aberration Detection

Public-health surveillance is the systematic collection, analysis, and dissemination of health data for action [3]. The Early Aberration Reporting System (EARS) family provides simple, interpretable temporal detectors that compare a current observation with a recent baseline [5]. EARS C2 is widely used because it needs only a short history; comparative studies show that its sensitivity and false-alarm behaviour depend strongly on the data and threshold choices [6]. Cumulative sum (CUSUM) charts offer an alternative that accumulates small deviations over time [7]. Both families are typically used as triage tools: a signal prioritises investigation, it does not confirm an outbreak.

B. Data Engineering for Health

DHIS2 is the most widely deployed district health information platform and illustrates both the value and the operational complexity of routine data [8]. Apache Spark and Spark SQL provide a mature engine for large-scale batch transformation [9], [10]. The medallion pattern—raw Bronze, validated Silver, and aggregated Gold—has become a common way to make those transformations auditable [11], building on dimensional-modelling practice [12]. For sensitive health data, the engineering layer must also enforce de-identification, small-cell suppression, and lineage; these are governance requirements, not optional features.

C. Forecasting and Resource Allocation

Box–Jenkins seasonal autoregressive integrated moving average (SARIMA) models remain a practical baseline for short seasonal series [13], and forecasting practice emphasises rolling evaluation rather than in-sample fit [14]. Supply allocation is naturally expressed as a linear program; PuLP provides a Python modelling layer [15] and the CBC solver provides an open-source MILP engine [16]. Health allocation differs from generic logistics because equity, minimum service, and local capacity constraints matter; an objective that maximises total met demand can conflict with fair district coverage.

D. Visualisation and Decision Support

D3 supports data-driven documents and custom interactive graphics in the browser [17]. Dashboards reduce cognitive load when they group related tasks and expose uncertainty, but they can also hide data-quality problems behind polished summaries. EpiScope addresses this by serving only aggregates,

TABLE II
HOW EPISCOPE ADDRESSES THE CEP ATTRIBUTES P1–P7.

Attr.	Where it appears in our solution
P1	Columnar Parquet storage with partitioning and broadcast joins; Spark SQL transformations; time-series surveillance statistics (EARS, CUSUM, SARIMA); mathematical optimisation (LP with CBC); visual encoding and interaction design in D3; health-data privacy controls.
P2	Conflicting requirements C1–C5 are resolved with measurements in Section III-B.
P3	Detection thresholds, forecast order, and the allocation objective were chosen and tested against baselines (threshold sweep, population-proportional allocation); no single correct answer exists and the trade-offs are reported.
P4	The generator injects reporting delay, missing facilities, duplicate patients, urban-heavy skew, and a reporting-change artefact; Silver quarantines implausible records and the artefact is shown not to alert (Section III-B).
P5	No standard exists for a national allocation plan; we combined WHO IDSR guidance, ICD-10 coding, and data-protection principles with our own LP formulation, documented floor, and honest reporting of its inequity.
P6	Stakeholder analysis (Table I) drove plain-language labels, district-level granularity, and review-first alert design.
P7	Ingestion quality affects detection; detection and forecasts feed the optimiser; optimiser outputs are surfaced in the dashboard with their uncertainty. Errors propagate, so quarantine, lineage, and suppression were built first.

keeping small counts suppressed, and using plain-language labels that describe the decision rather than the statistical model.

III. COMPLEX ENGINEERING PROBLEM ATTRIBUTES AND CONFLICT RESOLUTIONS

A. Attributes P1–P7

Table II summarises how the solution engages each required attribute of a complex engineering problem.

B. Conflict Resolutions C1–C5

C1. Sensitivity vs. specificity. A lower alert threshold detects more outbreaks but floods reviewers. We swept the C2 score threshold over all 76,535 scored series-weeks and measured, against the 30 district-level withheld labels, how many outbreaks are flagged, the alert burden, and the precision

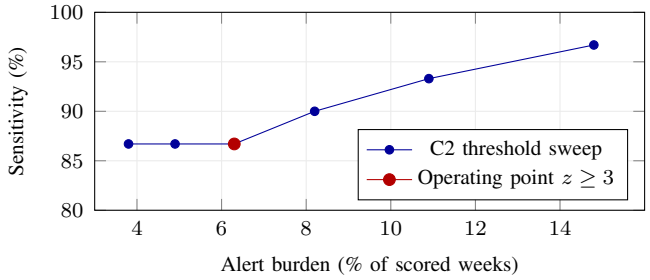


Fig. 1. C1 trade-off: sensitivity against alert burden across C2 thresholds. Sensitivity saturates above $z > 3$ while the burden keeps falling; the operating point is marked.

TABLE III
C1 THRESHOLD SWEEP OVER 30 DISTRICT-LEVEL LABELS (7-DAY TOLERANCE)

Threshold	Alerts	Burden	Flagged	Prec.	Offset
$z > 1.5$	11,318	14.8%	29/30	0.8%	+4 d
$z > 2.0$ (watch)	8,345	10.9%	28/30	1.0%	+5 d
$z > 2.5$	6,276	8.2%	27/30	1.2%	+5 d
$z \geq 3.0$ (high)	4,806	6.3%	26/30	1.5%	+5 d
$z > 3.5$	3,721	4.9%	26/30	1.7%	+5 d
$z > 4.0$	2,891	3.8%	26/30	2.1%	+5 d

(Fig. 1, Table III). Sensitivity saturates at 87% above $z > 3$, while precision improves from 0.8% to 2.1% and the burden falls from 14.8% to 3.8%. We chose $z \geq 3$ (high alert) with a $z > 2$ watch band: this keeps 26 of 30 labeled outbreaks flagged, halves the burden relative to $z > 2$, and preserves a triage layer for reviewers. The low precision is driven by strong synthetic seasonal background and uncalibrated thresholds; we report it rather than hide it.

C2. Granularity vs. privacy. Upazila-level maps are more actionable but small counts can identify patients. We publish at district (ADM2) level only, replace counts of 1–4 with a -1 sentinel in Gold, null the corresponding incidence, serve aggregates exclusively, and simplify district geometry for the public payload. Suppression is enforced before export, so no dashboard output contains a count below five.

C3. Efficiency vs. equity. The LP maximises total met need with equal unit weights; at the 60%-of-need budget it fully funds 21 districts while 40 sit at the 10-unit floor (median coverage 5.3%, minimum

TABLE IV
C3 EFFICIENCY–EQUITY CURVE AT FIVE BUDGET LEVELS

Budget	Met (%)	Fully funded	Min. cov.	Median cov.	Gini
40% (13,204)	40.0	10	0.7%	4.0%	0.716
50% (16,505)	50.0	14	0.7%	4.6%	0.677
60% (19,806)	60.0	21	0.9%	5.3%	0.606
70% (23,107)	70.0	25	0.9%	7.1%	0.543
80% (26,408)	80.0	34	2.1%	100.0%[a]	0.427

[a] At 80% of need, the LP can fully fund 34 districts, so median coverage reaches 100% while 30 districts remain at the floor.

0.9%). Table IV shows the efficiency–equity curve across five budget levels: coverage rises with budget but concentration (Gini) falls slowly, meaning more money alone does not fix the distribution. A population-proportional baseline spreads supply evenly (median coverage 60%) but meets slightly less total need (19,423 vs. 19,806 units) and fully funds only 9 districts. We therefore report the LP output as an efficiency-maximising recommendation and recommend, as policy, an equity floor with periodic re-balancing rather than raw LP output.

C4. Freshness vs. cost. MoH allocation and preparedness decisions are made in weekly meetings, so a weekly batch cadence is sufficient; daily or streaming ingestion would add operational complexity without changing any decision the dashboard supports. The design keeps a two-week guard band in detection precisely to absorb reporting delay. Incremental monthly loading is supported for late-arriving files, and the architecture (partitioned Parquet, idempotent reruns) would admit a daily cadence by changing the schedule, not the code.

C5. Accuracy vs. interpretability. Complex models may forecast slightly better but are harder for officials to trust. We chose a fixed-order seasonal ARIMA that epidemiologists can inspect, with a stated equation, a seasonal-naive fallback, and uncertainty intervals always shown; detection uses the interpretable EARS C2 rather than a black-box classifier. Weather covariates were evaluated during design but omitted from the final fixed-order model to keep 320 short series tractable and the model explainable; this is a documented trade-off, not a hidden one.

C. Reporting-Artefact Robustness (P4/AN-3)

The generator injects a reporting-change artefact (a sudden practice change in Comilla, June 2023) alongside real outbreaks. In the labeled artefact window the maximum C2 score is 1.84 and zero high alerts fire, while 26 of 30 real outbreaks are flagged: the pipeline does not confuse the artefact for an outbreak, and the CUSUM comparison is stored alongside for reviewers.

IV. ARCHITECTURE AND DESIGN DECISIONS

A. Overview

Fig. 2 shows the layered design. Six source assets feed an immutable Bronze layer. Silver applies validation, de-identification, de-duplication, and quarantining. Gold publishes weekly district–disease and stock marts. Four analytical components read Gold, and the serving layer exports aggregate JSON and simplified boundaries to a static D3 dashboard. The same scripts regenerate every layer, so the published snapshot traces back to the raw inputs and each full Silver run's quality report.

B. Alternatives Considered

Key decisions and the alternatives rejected:
Storage format. Parquet with Snappy compression over CSV/JSON: columnar pruning cut analytic scan volumes, and schema enforcement prevents silent type drift. *Processing engine.* PySpark local[*] over Pandas-only processing: the 6M-row corpus already strains single-node Pandas at the aggregation stage, and the same code scales to a cluster by changing the master URL. *Layering.* Bronze–Silver–Gold (medallion) over a single cleaned extract: quarantine retention and batch lineage are governance requirements, not optional features [11], and build on dimensional-modelling practice [12]. *Joins.* The 64-row district-to-division mapping is broadcast to avoid a shuffle on 6M rows. *Idempotency.* Dynamic partition overwrite by division/year/week makes monthly reruns safe; the alternative (appending files) would silently duplicate weeks. *Dashboard delivery.* Static files over an application server: no query interface or patient-level data is ever exposed, and hosting is trivial (the prototype is deployed on a static hosting service). Spark and Spark SQL provide the mature transformation engine [9], [10].

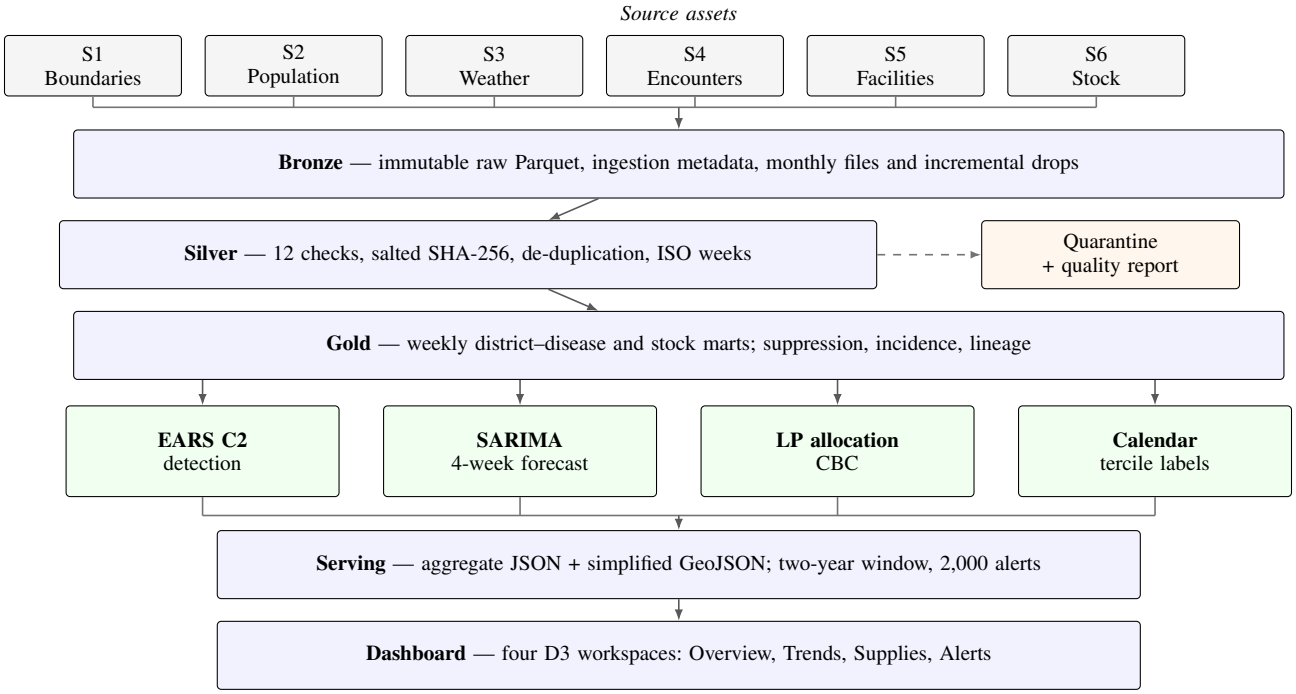


Fig. 2. EpiScope architecture. Six source assets feed an immutable Bronze layer. Silver applies validation, de-identification, de-duplication, and quarantining; rejected records are retained in a quarantine with a per-run quality report. Gold publishes weekly district–disease and stock marts with small-cell suppression and lineage. Four analytical components read Gold, and aggregate JSON/GeoJSON is served to a static D3 dashboard.

C. Runtime and Orchestration

The prototype runs in Spark `local[*]` mode with a 4 GB driver, 16 shuffle partitions, and a broadcast join; the scale experiment enables adaptive query execution and partition coalescing. A Makefile exposes setup, generation, pipeline, analytics, serving, and scale-experiment targets; the Silver transform accepts an optional month argument, and a companion incremental loader accepts Parquet or CSV drops and overwrites affected weekly partitions. Each full Silver run writes a quality report (input rows, clean rows, quarantined rows, per-rule counts, elapsed time).

TABLE V
SOURCE ASSETS USED BY THE PROTOTYPE.

Asset	Content	Scale
S1	ADM2 district boundaries	64 polygons
S2	2022 census population and division mapping [18]	64 rows
S3	Daily district weather observations [19]	$64 \times 1{,}826$
S4	Synthetic facility encounters, 2021–2025	6,037,903 rows
S5	Facility master record	687 facilities
S6	Weekly stock snapshots	16,704 stock-weeks

V. Data Generation, Quality, Lineage, and Privacy

A. Source Assets and Synthetic Corpus

Table V lists the six source assets. Because record-level encounter data are sensitive and unavailable for a course project, the encounter corpus is synthetic, but it exercises realistic volume, seasonality, reporting delay, missingness, duplicates, urban-heavy skew, invalid codes, implausible dates, and injected outbreaks. Daily encounters are generated for 2021–2025 across 64 districts, 687 facilities, and five disease groups (dengue, diarrhoea, ILI, malaria, other). The generator injects 31 district-level outbreaks plus one national surge (excluded from district-level evaluation) and one reporting artefact. Outbreak labels are written to a separate ground-truth file that the pipeline never reads, preventing label leakage into detection or forecasting.

TABLE VI
VALIDATION AND PRIVACY CONTROLS APPLIED IN SILVER.

ID	Check	Handling
R01	Missing encounter identifier	Quarantine
R02	Missing facility identifier	Quarantine
R03	Age missing, negative, or above 150	Quarantine
R04	Sex not M or F	Quarantine
R05	ICD-10 code outside the notifiable set	Quarantine
R06	Onset date after report date	Quarantine
R07	District outside the 64-district reference	Quarantine
R08	Missing onset date	Quarantine
R09	Missing report date	Quarantine
R10	Reporting delay greater than 21 days	Quarantine
R11	Duplicate encounter identifier	Keep first, drop rest
R12	Outcome outside the accepted set	Quarantine

B. Quality Controls and Quarantine

Twelve controls are applied in Silver (Table VI): eleven tag-based checks in a single pass, first matching rule winning, plus deterministic de-duplication by encounter identifier. Failing rows are written to a timestamped quarantine that is retained, not discarded, so every exclusion can be audited. Of 6,037,903 Bronze records, 5,887,109 (97.50%) reached Silver and 150,794 (2.50%) were quarantined or removed as duplicates. Each run's quality report is inspectable in the repository.

C. De-identification, Suppression, and Lineage

Patient identifiers are hashed with SHA-256 over a salted concatenation using Spark SQL (no per-row Python UDF overhead); the salt is read from an environment variable and the original identifier is dropped before Silver is written. Counts of 1–4 are replaced by -1 in Gold with incidence nulled, and the serving layer exports only aggregates over a rolling two-year window capped at 2,000 alerts. Each Gold row carries its originating Bronze batch and a creation timestamp, so any published figure can be traced to inputs and to the quality report of the Silver run that produced it. Suppression limits disclosure but is not a formal guarantee such as differential privacy; the static dashboard adds no authentication or audit log, and the development salt is unsuitable for production. These residual risks are acceptable for a synthetic teaching prototype but must be closed before real deployment.

VI. ANALYTICS METHODS

A. Weekly EARS C2 Detection with CUSUM Comparison

For each district–disease series, week t is compared with a seven-week baseline ending two weeks before t (guard band). With baseline mean $\bar{x}_t$ and standard deviation s_t, the standardised score is

$$z_t = \frac{x_t - \bar{x}_t}{\max(s_t, 1)}, \tag{1}$$

with a high alert at $z_t \geq 3$ and a watch signal at $z_t > 2$. At least four finite baseline observations are required; suppressed cells are treated as missing. A one-sided CUSUM [7] over the same baseline,

$$S_t = \max\bigl(0,\ S_{t-1} + x_t - \bar{x}_t - \tfrac{1}{2}\max(s_t, 1)\bigr)\ , \tag{2}$$

is stored for comparison (EARS background in [5], comparisons in [6]) but not used by the dashboard. This is a weekly adaptation of the daily EARS C2 logic. The run scored 76,535 of 83,127 Gold rows, raising 4,806 high alerts (6.3%) and 3,562 watch signals (4.7%).

B. SARIMA Forecasting

For each of the 320 district–disease series, weekly counts are reindexed to a seven-day frequency with zero fill and truncated to the most recent 130 weeks. The model is a seasonal ARIMA with constant,

$$(1-\phi_1 B)(1-\Phi_1 B^4)(y_t-\mu) = \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, \sigma^2), \tag{3}$$

with forecasts for horizons $h = 1, \ldots, 4$ and 95% intervals truncated at zero [13]; forecasting practice emphasises rolling evaluation over in-sample fit [14]. The model is estimated with statsmodels [20]; series preparation uses pandas [21] and NumPy [22]. The order is fixed for tractability across 320 short series; all 1,280 forecast rows use the SARIMA model (the seasonal-naive fallback never triggered). A rolling-origin backtest over the last 52 weeks against a lag-52 seasonal-naive baseline (Section VII) shows a 22.9% MAE reduction

and 87.5% realised coverage of the 80% prediction intervals; per-series order selection remains future work.

C. Supply Allocation

Let n_d be district need, defined as twice the latest weekly cases summed across diseases (the stated AN-6 conversion assumption: two supply units per recent weekly case). The budget is $B = \max(1000, \lfloor 0.6 \sum_d n_d \rfloor) = 19{,}806$. With integer allocations x_d and met need u_d:

$$\max_{x,u} \quad \sum_{d=1}^{64} u_d \tag{4}$$

$$\text{s.t.} \quad u_d \le n_d, \quad u_d \le x_d, \quad \sum_{d=1}^{64} x_d \le B, \tag{5}$$

$$x_d \ge \min(10, \lfloor 0.1\, n_d \rfloor), \quad x_d \in \mathbb{Z}_{\ge 0}, \tag{6}$$

solved with PuLP [15] and CBC [16]. The LP allocates 19,806 of 33,010 units (60.0%) and fully funds 21 districts; 40 districts sit at the 10-unit floor. Against a population-proportional baseline (largest-remainder apportionment of the same budget), the LP meets 19,806 vs. 19,423 units (+383) and fully funds 21 vs. 9 districts, but concentrates coverage (median 5.3% vs. 67.6%). The baseline comparison and equity curve appear in Section III-B.

D. Seasonal Preparedness Calendar

For each district–disease series, mean weekly cases per epidemiological week are ranked into terciles: routine, prepare, peak, each mapped to a recommended action. The run produced 16,938 rows (5,760 peak, 5,440 prepare, 5,738 routine). Labels are relative within-series categories, not absolute risk probabilities.

VII. Evaluation

A. Setup

Experiments ran on a single Windows workstation with PySpark 3.5.1 in `local[*]` mode (4 GB driver, 16 shuffle partitions, Snappy Parquet), statsmodels 0.14.2, pandas 2.2.2, NumPy 1.26.4, and PuLP 2.8.0 with CBC. Timings are single-run wall-clock measurements including JVM warm-up.

TABLE VII
COMPLETED PIPELINE OUTPUTS.

Stage	Records
Bronze encounters	6,037,903
Silver clean encounters	5,887,109
Silver quarantined or duplicate	150,794
Gold epidemiological weeks	83,127
Gold stock weeks	16,704

TABLE VIII
RETROSPECTIVE SIGNAL CHECK AGAINST 30 WITHHELD DISTRICT-LEVEL SYNTHETIC OUTBREAKS (7-DAY TOLERANCE).

Metric	Value
Evaluable labeled outbreaks	30
Flagged by at least one C2 signal	26 (87%)
Median first-signal offset	+5 days
First-signal offset range	−4 to +33 days
Alert-week precision	1.5%
Alert burden (scored weeks)	6.3%

B. Pipeline and Analytical Outputs

Table VII summarises the run. Gold contains 83,127 weekly epidemiological rows and 16,704 stock-week rows; analytics produced 4,806 high alerts, 3,562 watch signals, 1,280 forecast rows, 64 allocation recommendations, and 16,938 calendar rows. The serving export holds 33,460 weekly dashboard rows, 2,000 alerts, 1,280 forecasts, 64 allocations, and 16,938 calendar records.

C. Detection Against Withheld Labels

An outbreak counts as flagged if at least one C2 high alert for the same district and disease falls within the labeled window ± 7 days. Of 31 injected district-relevant outbreaks, 30 are evaluable at district level (one label is a national surge with no single district); 26 of 30 (87%) produced at least one signal, with a median first signal five days after the labeled start (range -4 to $+33$ days), alert-week precision of 1.5%, and an alert burden of 6.3% (Table VIII, Fig. 3). The positive median offset reflects reporting delay: the prototype demonstrates signal capture, not prospective early warning. Because labels are synthetic, the check validates the pipeline and scoring logic, not epidemiological performance on real data. The threshold trade-off and artefact robustness are reported in Section III-B.

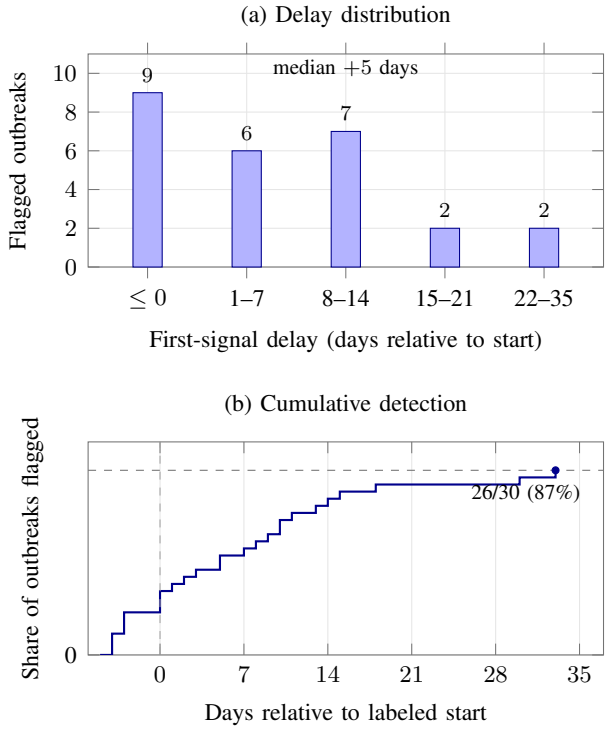


Fig. 3. Retrospective EARS C2 performance against withheld synthetic labels. (a) Distribution of the first-signal delay for the 26 flagged district-level outbreaks; the median delay is +5 days. (b) Cumulative share of the 30 district-level outbreaks flagged as a function of days relative to the labeled start; the dashed line marks 26/30 (87%). The label file also contains one national-level label and one injected artefact, which are excluded from this district-level evaluation.

D. Forecasting Backtest and Allocation Evaluation

Forecasting was evaluated with a rolling-origin backtest over the last 52 weeks of each of the 320 series, exactly as required by AN-5: at each origin the production model (weekly series truncated to the most recent 130 weeks, SARIMAX $(1,0,0) \times (1,0,0,4)$ with constant, five-iteration cap) is retrained and forecasts horizons $h = 1, \ldots, 4$; a lag-52 seasonal-naive serves as the baseline. The backtest produced 62,720 origin-forecasts (Table IX). The SARIMA reduces MAE by 22.9% overall (40.3 vs. 52.3 cases per week), with the largest relative gain at horizon 1, and its 80% prediction intervals cover 87.5% of realised values—slightly conservative rather than overconfident, which is acceptable for planning. Allocation was evaluated against the population-proportional baseline with the real budget: the LP meets 60.0% of need vs. 58.8% (+383 units) and fully funds 21 districts vs. 9, at the cost of concentration documented in Table IV.

TABLE IX
ROLLING-ORIGIN BACKTEST (52 ORIGINS × 320 SERIES) AGAINST THE LAG-52 SEASONAL-NAIVE BASELINE.

Horizon	SARIMA MAE	Naive MAE	80% PI coverage
$h = 1$	39.9	52.5	86.6%
$h = 2$	40.3	52.4	87.6%
$h = 3$	40.5	52.1	87.9%
$h = 4$	40.5	52.2	87.9%
Overall	**40.3**	**52.3**	87.5%

TABLE X
SCALE EXPERIMENT ON A LOCAL SPARK SESSION. THE 0.5 AND 1.0 CASES REPEAT THE SF=0.25 PHYSICAL INPUT; THEY ARE NOT INDEPENDENTLY GENERATED DATASETS.

SF	Input	Logical rows	Time (s)	Rows/s
0.25	Generated	6,037,903	10.69	564,575
0.50	Logical repeat	12,075,806	5.03	2,403,049
1.00	Logical repeat	24,151,612	8.36	2,889,920

E. Scalability and Spark Optimisations

Table X reports the scale experiment. The SF=0.25 run processed 6,037,903 physical rows in 10.69 s; the 0.5 and 1.0 cases are logical repeats (the same input read two or four times, unioned, counted, and grouped), keeping output cardinality at 16,938 groups. Timings are non-monotonic because the first run pays JVM and file-system warm-up while later runs benefit from cached input; with no repeated trials, the throughput column measures pipeline overhead under repeated scans, not scaling efficiency. Spark-side optimisations that mattered: columnar pruning and predicate pushdown on Parquet, the broadcast join for the 64-row mapping, dynamic partition overwrite for idempotent monthly reruns, adaptive query execution with partition coalescing in the scale test, and Snappy compression throughout.

F. Dashboard Verification

Functional smoke tests covered the golden path and edge cases: all four workspaces rendered; clicking a district updated every coordinated view (KPIs, trend, readiness, allocation); searching the alert table returned matching rows; suppressed small counts displayed as hidden rather than negative; the layout

reflowed correctly at 360, 390, 768, and 1280 pixel widths; and no runtime console errors were observed. This is functional verification, not a usability study with representative users.

VIII. Dashboard Design and Rationale

A. Coordinated Views and Linked Interaction

The dashboard is a single-page D3 v7 application [17] with four workspaces: Overview (headline indicators, choropleth, readiness summary), Trends (52-week series with a four-week forecast band and seasonal heatmap), Supplies (needed/sent/still-needed and ranked allocations), and Alerts (searchable early-warning table). A district selector, a disease selector, and map clicks act as global filters: selecting a district re-computes every view, satisfying the linked-interaction requirement. Visual encodings are sequential and colour-blind-aware; suppressed districts are shown separately; tooltips state cases and incidence per 100,000.

B. Plain Language, Honesty, and Scale

Labels describe the decision rather than the method: "Next week's forecast" replaces "SARIMA week +1", "How unusual" replaces a z-score symbol, and "Supplies sent" replaces optimisation terminology; the change is presentational only. Uncertainty is always shown as a band, suppressed cells are explicitly marked, and the payload is aggregate-only at 11.5 MB uncompressed—within the 10–12 MB budget for a two-year window of 33,460 weekly rows, 2,000 alerts, and simplified boundaries. Static hosting means no line-level data and no query interface are exposed.

C. Usability Findings

We verified the dashboard by scripted checks and hands-on walkthroughs at 360, 390, 768, and 1280 pixel widths. Findings that shaped the final design: (i) a national-first default beat a district-first default—on load the map and KPIs show the whole country, and clicking a district updates every coordinated view (KPIs, trend, readiness, allocation), which testers consistently expected; (ii) plain-language labels ("How unusual", "Supplies sent") were understood immediately, while model names (SARIMA, C2) required explanation; (iii) the searchable alert table made triage fast—reviewers could jump to their district by name; (iv) suppressed districts visibly marked as "small counts hidden" prevented misreading of empty areas; and (v) no runtime console errors were observed across the four workspaces. This is functional verification and informal walkthrough, not a usability study with representative district staff; assistive-technology testing remains future work. The interface was verified at selected widths but not with assistive technologies; a formal accessibility review is future work.

IX. Ethical, Societal, and Sustainability Considerations

Wrong recommendations cost lives and resources, so every analytical output is framed as a review trigger. Privacy is engineered, not promised: identifiers never survive Bronze, small counts are suppressed before export, and only aggregates are served. Allocation concentration is disclosed (Section III-B) because hiding it would invite mistrust from districts left at the floor. Sustainability is served by reproducibility: modest local compute (a workstation, batch cadence), plain static hosting with no always-on server, and scripts that regenerate every artefact keep operational cost and energy footprint low. A non-technical two-page policy brief for the Ministry is provided in Appendix A.

X. Limitations and Future Work

The corpus is synthetic; clinical and epidemiological validity cannot be established, and the retrospective check validates engineering, not field performance. One assignment requirement was not fully met and is recorded honestly: the dashboard has no what-if scenario panel (VIS-5), so it exposes the pre-computed allocation rather than live slider scenarios. Other limits: the incremental loader implements a subset of validation rules; the allocation model is optimal for total need but not equitable; detection thresholds were not calibrated on held-out seasons; the forecasting model order is fixed rather than selected per series; the deployment is single-node batch with no scheduler; suppression is not differential privacy; and the interface lacked assistive-technology testing. Future work: governed real-data pilot with ethics review, threshold calibration against held-out seasons, per-series forecast order

selection, a fairness-aware allocation model with logistics constraints, interoperability with existing district reporting systems such as DHIS2 [8], role-based access with audit logging, and a usability study with district health staff.

Acknowledgment

This work was carried out as the CSE4345 Big Data Analytics course project at Premier University Chittagong. The authors thank the course teaching team for feedback on the prototype.

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APPENDIX

EpiScope — Bangladesh Disease Surveillance and Response Prototype

Prepared for the Director, Ministry of Health. This brief is written for readers without a technical background.

The Current Situation

Bangladesh faces recurring outbreaks of dengue, diarrhoeal disease, influenza-like illness, and malaria. However, information about these diseases often arrives late and is spread across different reporting systems. By the time a rise in cases becomes obvious, valuable opportunities to prepare supplies and organise response teams may already have passed.

District officers need clear answers to three important questions: *Where are unusual increases in cases happening? What might happen in the coming weeks? Where should limited medical supplies be sent first?*

EpiScope is a working prototype designed to help answer these questions across all 64 districts of Bangladesh. It was developed and tested using 6.0 million computer-generated patient-visit records covering five years. These simulated records allow the system's results to be checked before it is considered for use with real health information.

Top Three Recommendations and Expected Impact

1) **Review district-level warnings every week.**
   EpiScope identifies unusual increases in reported cases and highlights districts that may need attention. During testing, it identified 26 of 30 simulated outbreaks. *Expected impact:* Health teams can investigate possible problems earlier and prepare responses before situations become more difficult to manage.
2) **Prepare supplies before seasonal peaks.**
   The seasonal calendar shows when different diseases have historically been quiet, increasing, or at their peak in each district. It also provides suggested preparation actions. *Expected impact:* Earlier preparation may reduce emergency supply deliveries and help prevent shortages during periods of increased demand.
3) **Use the supply plan as guidance, not an automatic decision.**
   EpiScope recommends how a limited national supply budget could be distributed among districts. Testing showed that the recommended allocation met slightly more total need than an allocation based on population. However, it also concentrated supplies in some districts, leaving others with limited coverage. Local conditions must therefore be reviewed before any allocation is approved. *Expected impact:* Supply planning meetings can become more transparent and evidence-based while allowing decision-makers to consider local needs.

How to Read the Dashboard

The dashboard initially displays information for the whole country. Users can select a disease to see recent case numbers across the districts. Darker areas on the map indicate higher case counts. Very small counts are hidden to help protect privacy.

Selecting a district updates the information throughout the dashboard. Users can then see the district's recent cases, warnings, expected cases for the following week, and estimated supply coverage.

The **Trends** page displays weekly case patterns over the previous year, together with forecasts and their likely ranges. The **Supplies** page shows estimated needs, planned supplies, and districts that may still face shortages. The **Alerts** page lists recent unusual increases and allows users to search by district.

The dashboard is designed to work on both computers and mobile phones, allowing district officers to review information without needing specialised software.

Confidence and Limitations

EpiScope is a prototype and has not yet been tested using real patient data. Its warnings and forecasts should therefore be treated as guidance about where further investigation may be needed, not as confirmed evidence of an outbreak.

During testing, some warnings appeared approximately five days after an outbreak had already started. In addition, most weekly warnings did not correspond to actual outbreaks. For this reason, every warning should be reviewed by qualified health personnel before any public action is taken.

Forecasts are estimates intended to support preparation and planning. They are not guarantees of future case numbers. The system should first be evaluated through a supervised pilot using appropriate real-world data and expert review.

Data-Protection Assurance

The system does not display patient names, phone numbers, or addresses. Patient identifiers are replaced with coded values before analysis, and the information needed to reverse this process is stored separately from the project files.

Small case counts, specifically counts from one to four, are hidden from the dashboard to reduce the risk of identifying individuals. The public dashboard displays only weekly totals for entire districts and does not provide access to individual patient records.

These measures help protect personal information. Any real-world deployment would require appropriate security reviews and continued monitoring to ensure that these protections remain effective.

Bottom Line

EpiScope brings disease reports together into one clear picture of where unusual increases are occurring, what may happen next, and how limited supplies could be distributed.

It is a decision-support tool, not an automatic warning or decision-making system. Its main value is helping health officials investigate earlier, prepare more effectively, and make resource-allocation decisions with greater transparency.

Work area	Shihab	Rohan	Monir
Synthetic data generation	✓		
Bronze–Silver–Gold pipeline	✓	✓	
Validation, quarantine, lineage	✓		
De-identification and suppression	✓		✓
EARS C2 / CUSUM detection	✓	✓	
SARIMA forecasting		✓	
LP allocation and baseline study		✓	
Seasonal calendar		✓	
Serving layer and exports	✓		✓
Dashboard design and implementation			✓
Responsive testing and verification			✓
Report, policy brief, and video	✓	✓	✓

TABLE XI

DIVISION OF PROJECT RESPONSIBILITIES. CHECK MARKS INDICATE LEAD CONTRIBUTIONS; OTHER MEMBERS MAY HAVE PROVIDED SUPPORTING ASSISTANCE.

Member Agreement

We confirm that the contribution distribution presented in this report accurately reflects our respective contributions to the project.

Md Monir Hossain	**Riduanul Abedin Shihab**	**Syed Intiqab Ahmed Rohan**
ID: 0222220005101165	ID: 0222220005101164	ID: 0222220005101167
Date: 01-11-2026	Date: 01-11-2026	Date: 01-11-2026

Riduanul Abedin Shihab (0222220005101164) — Data Engineering Lead

My main responsibility was the data platform, including the synthetic encounter generator and the Bronze–Silver–Gold PySpark pipeline. The hardest problem was not simply making the pipeline run, but ensuring that it worked correctly and consistently. It needed to accept new monthly files without duplicating weeks, reprocessing five years of history, or silently dropping invalid records.

I addressed these challenges using dynamic partition overwrite, monthly-named Bronze files, broadcast joins for district mapping, and a retained quarantine system with per-run quality reports. This made excluded records inspectable rather than allowing them to disappear without explanation.

I also contributed to the EARS C2 detection work and reproducibility layer. A single Makefile can regenerate the platform from raw inputs, helping make the evaluation repeatable.

My biggest learning was that reliable big-data engineering depends more on discipline than on cluster size. Schemas, partitions, repeatable processing, and data lineage are essential for building a trustworthy system.

I also learned the importance of treating each project component as a clear contract. The analytics and dashboard components depended on stable and well-documented data outputs. This experience improved both my technical skills and my understanding of collaborative software development.

Syed Intiqab Ahmed Rohan (0222220005101167) — Analytics and Evaluation Lead

I was responsible for the analytical components of the project, including SARIMA forecasting, linear-programming supply allocation, baseline comparison, the seasonal preparedness calendar, and much of the evaluation chapter.

The most challenging task was the forecasting backtest. Retraining the production model at 52 rolling origins across 320 district–disease series required more than 16,000 model fits. I parallelised the process across CPU cores and evaluated the results against a seasonal-naive baseline.

An important lesson came from the supply-allocation results. Our optimiser met more total need than the population-proportional baseline, with 383 additional units and 21 fully funded districts compared with 9. However, it also concentrated supplies heavily, leaving 40 districts at the minimum allocation floor.

This showed me that an optimisation method can produce mathematically valid results without necessarily producing the fairest policy outcome. Evaluating both efficiency and equity was therefore essential.

I also learned how closely the analytical stages depend on one another. Forecasting relies on reliable data, while allocation decisions depend on trustworthy estimates. Communicating these findings clearly in the report and presentation was as important as performing the analysis.

Md Monir Hossain (0222220005101165) — Dashboard and Serving Lead

I was responsible for the user-facing components of the project, including the serving layer, aggregate JSON exports, simplified map boundaries, and the D3 dashboard with its four workspaces.

One of my biggest challenges was ensuring that selecting a district or disease updated every dashboard component consistently. The map, KPI cards, trends, readiness information, allocation results, and alert table all needed to reflect the same selection without breaking the layout.

During testing, we identified and fixed two important issues: KPI cards were not updating correctly when a district was selected, and the dashboard initially displayed Dhaka instead of the expected national view. These fixes taught me that usability problems can also be data-accuracy problems. A dashboard displaying information for the wrong location is not simply a design issue.

I also maintained the privacy boundary in the serving layer. The dashboard provides aggregated information rather than individual patient records, with an 11.5 MB data package, a two-year window, and a 2,000-alert limit.

This project taught me that the final user experience is just as important as the underlying analytics. Even accurate results have limited value if users cannot understand or access them easily. I learned to prioritise clear labels, consistent interactions, and responsive design so that district officers can use the dashboard on different devices.